

Won Hwa Kim
#4412 RIST Building IV, POSTECH,
77 Cheongam-ro, Nam-gu, Pohang, South Korea

RESEARCH INTERESTS

My research is focused on various topics in **Machine Learning**, **Computer Vision** and **Medical Imaging**. I am particularly interested in applied harmonic analysis in non-Euclidean spaces (e.g., signal processing on graphs) and stochastic process (e.g., longitudinal analysis) to develop efficient Deep Learning frameworks to facilitate understanding of neurodegenerative disorders such as Alzheimer's Disease (AD) and Dementia.

APPOINTMENTS

Associate Professor	2023 - present
Assistant Professor	2020 - 2023
Computer Science and Engineering, POSTECH, S. Korea	
Graduate School of Artificial Intelligence, POSTECH, S. Korea	
Medical Science and Engineering, POSTECH, S. Korea	
 Assistant Professor	 2018 - 2023
Computer Science and Engineering, University of Texas at Arlington, U.S.A.	
 Researcher	 2017 - 2018
Data Science Team, NEC Labs., America, U.S.A.	
 Research Engineer	 2010 - 2011
Environmental Tech Center, Hyundai Motors Company, S. Korea	
 Information Management Officer	 2003 - 2005
Headquarter, ROK Army Support Group, the 8th U.S. Army Division, S. Korea	

EDUCATION

University of Wisconsin - Madison , Madison, Wisconsin, U.S.A.	2011 - 2017
Ph.D., Computer Sciences (Minor in Statistics)	
• Thesis: A Multi-resolution Framework for Statistical Analysis of Neuroimaging Data	
• Advisor: Vikas Singh	
 KAIST , Daejeon, South Korea	 2008 - 2010
M.S., Robotics Program	
• Thesis: Diversified Emotions with Mood for Human-like Behaviors of Robots	
• Advisor: Myungjin Chung	
 Sungkyunkwan University , Seoul, South Korea	 2001 - 2008
B.S., Electrical Engineering (<i>Early graduation in 7 semesters</i>)	

HONORS and AWARDS

• <i>2 Outstanding Paper Awards</i> (Silver and Bronze), IPIU	2026
• <i>Best Paper Award (with Nvidia DGX Spark)</i> , MICCAI GRAIL Workshop	2025
• <i>Outstanding Healthy Laboratory</i> (Excellence Award), POSTECH Human Right Center	2025
• <i>Excellent Invention Award</i> (Excellence Award), SK Hynix	2025
• <i>Best Editor Award</i> , ICT Express	2025
• <i>2 Outstanding Poster Awards</i> (Excellence Award and Encouragement Award), KSIIM Conference	2025
• <i>Outstanding Poster Award</i> , IPIU	2025
• <i>Top-10 Healthy Laboratory</i> (Encouragement Award), POSTECH Human Right Center	2024
• <i>First Place Award</i> , UWF4DR Challenge in MICCAI	2024
• <i>Outstanding Paper Award</i> (Bronze), IPIU	2024
• <i>3 Outstanding Paper Awards</i> (Silver, Bronze, Encouragement), IPIU	2023

- *NSF CISE CAREER Workshop Travel Award*, National Science Foundation (NSF) 2019
- *Rising STARS Award*, University of Texas System [\$250,000] 2017
- *Doctoral Consortium Travel Award*, Computer Vision and Pattern Recognition (CVPR) 2016
- *Student Travel Award*, Medical Image Computing and Computer Assisted Intervention (MICCAI) 2013
- *Machine Learning Summer School (MLSS) Scholarship*, University of California, Santa Cruz 2012
- *National Fellowship*, S. Korea 2008 - 2010
- *Finalist for Best Paper in Biomimetics*, International Conference on Robotics and Biomimetics 2009
- *Merit Based Scholarship*, Sungkyunkwan University 2002, 2003, 2005
- *3rd Place in 12th Grade*, Utah Math Contest 2001

GRANTS

- (RS-2026-25616313) Trustworthy Agentic AI for Medical Image Analysis, 2026 - 2031
AI Star Fellowship, Institute for Information and Communication Technology Planning and Evaluation (IITP),
Role: **PI**, [₩11B (~\$8M)]
- (RS-2026-25494850) Physics-Informed High-Order Graph Foundation Model for Counterfactual 2026 - 2028
Prediction of Neurodegenerative Diseases,
Core Research Grant (Merit Research Program), National Research Foundation (NRF),
Role: **PI**, [₩1,050M (~\$800,000)]
- Multi-task Graphic Design System Foundation Model, 2025 - 2030
Strategic Industry-Academia Project, Samsung DS,
Role: **PI**, [₩400M (~\$308,000)]
- (RS-2025-02216257) Geometric Deep Learning for Unstructured Neuroimage Analysis, 2025 - 2028
Basic Research Lab (BRL), National Research Foundation (NRF),
Role: **PI**, [₩1,500M (~\$1.15M)]
- Research Group for Next-Generation Strategies in Brain Glymphatic and Neural Regulation, 2025 - 2030
Global Top Strategical Center, Ministry of Science and ICT,
Role: **Sub-project Investigator**, [Total: ₩25,000M (~\$19.2M)]
- Fundus Image Foundation Model, Mediwhale 2025
Role: **PI**, [₩30,000,000 (~\$22,000)]
- Graph Foundation Model, Samsung Advanced Institute of Technology (SAIT), 2024 - 2026
Role: **PI**, [₩200M (~\$154,000)]
- Consulting on VUNO MED Solution for Advanced Research, VUNO, 2023 - 2024
Role: **PI**, [₩12,000,000 (~\$10,000)]
- Solder Joint Failure Detection Algorithm Development, SK Hynics (via AICC), 2023
Role: **PI**, [₩50,000,000 (~\$42,000)]
- (RS-2022-NR070486) Developing Graph Deep Learning Framework for Analysis for Analysis of 2022 - 2026
Early Diagnosis and Symptoms of Alzheimer's Disease via Brain Connectome,
Mid-career Researcher Program, National Research Foundation (NRF),
Role: **PI**, [₩540M (~\$450,000)]
- (HU22C016800) Development of a K-dementia Bigdata Central HUB Database and 2022 - 2024
Analysis Platform (with Dr. Hwang at SNU),
Korea Health Industry Development Institute (KHID),
Role: **PI**, [₩401M (~\$334,000)]
- (IITP-2202-0-00290) Visual Intelligence for Space-Time Understanding and Generation 2022 - 2026
based on Multi-layered Visual Common Sense,
Institute for Information and Communication Technology Planning and Evaluation (IITP),
Role: **Co-PI**, [Total: ₩3,950M (~\$3.3M)]
- (NIH R03 AG070701) Continuing Tool Development for Longitudinal Network Analysis: 2021 - 2023
Enriching the Diagnostic for Longitudinal Network Analysis: Enriching the Diagnostic Power of
Disease-Specific Connectomic Biomarkers by Deep Graph Learning (with Dr. Wu at UNC-Chapel Hill),
Small Research Grant, National Institute of Health (NIH),
Role: **Co-I**, [UTA: \$125,353]

- (NSF IIS CRII 1948510) Learning Novel Multi-resolution Representations of Graphs: Applications to Brain Connectivity Analysis for Alzheimer’s Disease, *Computer and Information Science and Engineering Research Initiation Initiative (CRII)*, National Science Foundation (NSF), Role: **PI**, [\$175,000] 2020 - 2022
- (NSF IIS SMALL 2008602) An Optimization Framework for Designing Derived Attributes with Humans-in-the-loop (joint work between UTA and NJIT), *NSF CORE:Small*, National Science Foundation (NSF), Role: **Co-PI**, [Total: \$498,762] 2020 - 2022
- (NIH R01 AG059312-01A1) Algebraic Formulations for Characterizing Structural Brain Connectivity Changes and Pathology Transmission Networks in Preclinical Alzheimer’s Disease (with Dr. Singh at UW-Madison), National Institute of Health (NIH), Role: **Co-I**, [UTA: \$150,785] 2019 - 2021
- (IITP-2020-2015-0-00742) High-Potential Individuals Global Training Program (with SKKU) Institute for Information and Communications Technology Promotion (IITP), Role: **PI**, [\$33,034] 2019 - 2020
- Convolution Neural Network for Graph Data, *Research Enhancement Program (REP)*, University of Texas at Arlington, Role: **PI**, [\$10,000] 2018 - 2019
- (CTEDD 018-08) Social Media Analysis for Transportation Assessment (with Georgia Tech), *Center for Equity, Diversity and Dollar (C-TEDD)*, US Department of Transportation (USDOT), Role: **PI**, [\$101,933] 2018 - 2019

PUBLICATIONS

Note: Top-tier conferences in computer science are valued as prestigious journals in other areas.

1. Protay Dey, Ayush Roy, Hyuna Cho, **Won Hwa Kim**, Vishnu Suresh Lokhande, “MAGNET: Manifold-Aware Graph Diffusion Network for Connectome Generation”, *Neural Information Processing Systems (NeurIPS)*, 2026.
2. Serim Ryou, Jayeon Yoo, Minjae Jeong, **Won Hwa Kim**, Seong-Jin Park, “ChipLayoutBench: A Multi-modal Benchmark for Vision-Language Models in Semiconductor Layout Understanding”, *Empirical Methods in Natural Language Processing (EMNLP Findings)*, 2026.
3. Seunghun Baek, Jihwan Park, Jaeyoon Sim, Minjae Jeong, Hoseok Lee, **Won Hwa Kim**, “Residual-Guided Expert Specialization for Incomplete Multimodal Learning”, *European Conference on Computer Vision (ECCV)*, 2026.
4. Jaeyoon Sim*, Hoseok Lee*, Jihwan Park, Seunghun Baek, Yu Zhang, **Won Hwa Kim**, “Spectral-Temporal State Space Modeling on Functional Brain Networks”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026. [**Spotlight**: ~4.63%, **Provisional accept**: ~9%, *: equal contribution]
5. Minjae Jung*, Seungjoo Lee*, Jaejin Lee, **Won Hwa Kim**, “Learning Robust Medical Image Segmentation under Mixed-Quality Annotation”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026. [**Provisional accept**: ~9%, *: equal contribution]
6. Yujee Song*, Seunghun Baek*, Guorong Wu, **Won Hwa Kim**, “ENC-ODE: Event-level Neurodegenerative Modeling in Continuous Time with Neural ODEs”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026. [*: equal contribution]
7. Minjae Jeong*, Jaejin Lee*, Yechan Hwang, **Won Hwa Kim**, “SurgMark: Structure-Aware Watermarking for 4D Gaussian Splatting in Surgical Scenes”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026. [*: equal contribution]
8. Seunghun Baek*, Jihwan Park*, Jaeyoon Sim, Hoseok Lee, Jihwan Park, Seungjoo Lee, **Won Hwa Kim**, “Set-Inclusive Uncertainty Modeling for Robust Brain Tumor Segmentation”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026. [*: equal contribution]
9. Junho Choi, Mingyu Joo, Jiwon Son, **Won Hwa Kim**, Ilwoo Lyu, “SPHARM-Mamba: Rotation-Invariant Multiscale Modeling for Brain Age Prediction”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026.

10. Minhee Lee, Sangyoon Lee, Jiwook Lee, Minki Hong, Kyuyoung Kim, **Won Hwa Kim**, Jaeho Lee, “Multi-frame Restoration for High-rate Lissajous Confocal Laser Endomicroscopy”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026.
11. Tingting Dan, Zhixuan Zhou, Won **Won Hwa Kim**, Guorong Wu, “A Variational Mean-Field Control Framework for Graph Representation Learning”, *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2026.
12. Jaeyoon Sim, Soojin Hwang, Seunghun Baek, Guorong Wu, **Won Hwa Kim**, “Learning Multi-Scale Hypergraph for High-Order Brain Connectivity Analysis”, *International Conference on Machine Learning (ICML)*, 2026.
13. Junhyoung Chung, Euijong Song, **Won Hwa Kim**, Gunwoong Park, “Convex Distance Operator Transport: Convex and Geometry-Preserving Information”, *International Conference on Machine Learning (ICML)*, 2026.
14. Minjae Lee*, Sungwoo Hur*, Soojin Hwang, **Won Hwa Kim**, “PR-MaGIC: Prompt Refinement Via Mask Decoder Gradient Flow For In-Context Segmentation”, *Computer Vision and Pattern Recognition (CVPR)*, 2026 [**Oral presentation**, *: equal contribution]
15. Jaejin Lee*, Minjae Jeong*, Joonhyuk Park, Yechan Hwang, Seunghun Baek, **Won Hwa Kim**, “Mark4D: Temporally-Consistent Watermarking for 4D Gaussian Splatting”, *Computer Vision and Pattern Recognition (CVPR)*, 2026 [*: equal contribution]
16. Sourav Pal, Viet Luong, Hoseok Lee, Tingting Dan, Guorong Wu, Richard Davidson, **Won Hwa Kim**, Vikas Singh, “MnemoDyn: Learning Resting State Dynamics from 40K fMRI sequences”, *International Conference on Representation Learning (ICLR)*, 2026
17. Ayush Roy, Samin Enam, Jun Xia, **Won Hwa Kim**, Vishnu Suresh Lokhande, “Is Exchangeability better than I.I.D. to handle Data Distribution Shifts while Pooling Data for Data-scarce Medical image segmentation?”, *Medical Imaging with Deep Learning (MIDL)*, 2026
18. Soojin Hwang*, Jaeyoon Sim*, **Won Hwa Kim**, “HiMix : Hierarchical Visual-Textual Mixing Network for Lesion Segmentation”, *Winter Conference on Applications of Computer Vision (WACV)*, 2026 [*: equal contribution]
19. George R. Nahass, Zhu Wang, Homa Rashidisabet, **Won Hwa Kim**, Sasha Hubschman, Jeffrey C. Peterson, Pete Setabutr, Chad A. Purnell, Ann Tran, Darwin Yi, Sathya N. Ravi, “Targeted Unlearning Using Perturbed Sign Gradient Methods With Applications On Medical Images”, *Transactions on Machine Learning Research (TMLR)*, 2025.
20. Dougho Park, Jinmi Kim, Da Woon Kim, Donghyun Lee, Taeyeon Kim, Dahyeon Koo, Youjin Lee, **Won Hwa Kim**, Hyo Jin Kim, “Dietary Patterns Associated with the New Onset of Chronic Kidney Disease Using Clustering Algorithm”, *Scientific Reports*, 2025.
21. Joonhyuk Park*, Donghyun Lee*, Yujee Song, Guorong Wu, **Won Hwa Kim**, “Topology-aware Graph Diffusion Model with Persistent Homology”, *Neural Information Processing Systems (NeurIPS)*, 2025. [*: equal contribution]
22. Tingting Dan, Xinwei Huang, **Won Hwa Kim**, Guorong Wu, “Explore In-Context Message Passing Operator for Graph Neural Networks in A Mean Field Game”, *Neural Information Processing Systems (NeurIPS)*, 2025.
23. Seunghun Baek, Jaejin Lee, Jaeyoon Sim, Minjae Jung, **Won Hwa Kim**, “MNM: Multi-level Neuroimaging Meta-analysis with Hyperbolic Brain-Text Representations”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025. [**Provisional accept:** ~9%]
24. Hyuna Cho, Hayoung Ahn, Guorong Wu, **Won Hwa Kim**, “Adaptive Adversarial Data Augmentation with Trajectory Constraint for Alzheimer’s Disease Conversion Prediction”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025. [**Provisional accept:** ~9%]
25. Joonhyuk Park, Donghyun Lee, Guorong Wu, **Won Hwa Kim**, “Conditional Graph Diffusion with Topological Constraints for Brain Network Generation”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025.
26. Wooseok Jung*, Joonhyuk Park*, **Won Hwa Kim**, “DISCLOSE the Neurodegeneration Dynamics: Individualized ODE Discovery for Alzheimer’s Disease Precision Medicine”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025. [**Oral presentation:** ~2.2%, *: equal contribution]
27. Sungyoon Jung, Donghyun Lee, **Won Hwa Kim**, “MindLink: Subject-agnostic Cross-Subject Brain Decoding Framework”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025.

28. Soojin Hwang, Jaeyoon Sim, **Won Hwa Kim**, “REMix: Refinement-Enhanced Visual-Textual Mixing for Lesion Segmentation”, *MICCAI Workshop on Emerging LLM/LMM Applications in Medical Imaging (ELAMI)*, 2025.
29. Hayoung Ahn, Seungjoo Lee, Jaeyoon Sim, Yechan Hwang, Hyuna Cho, Guorong Wu, **Won Hwa Kim**, “Spectral Graph Autoregressive Modeling for Conditional Brain Network Augmentation”, *MICCAI Workshop on GRaphs in biomedicAl Image anaLysis (GRAIL)*, 2025.
30. Hyuna Cho, Ziquan Wei, Seungjoo Lee, Tingting Dan, Guorong Wu, **Won Hwa Kim**, “Conditional Diffusion Model using Ordinal Regression for Longitudinal Neurodegenerative Data Generation”, *Alzheimer’s Association International Conference (AAIC)*, 2025.
31. Sehyung Cheong, Hoseok Lee, **Won Hwa Kim**, “Survey on Sampling Conditioned Brain Images and Imaging Measures with Generative Models”, *Biomedical Engineering Letters (BMEL)*, 2025.
32. Inho Kim, Youngkil Song, Jicheol Park, **Won Hwa Kim**, Suha Kwak, “Improving Sound Source Localization with Joint Slot Attention on Image and Audio”, *Computer Vision and Pattern Recognition (CVPR)*, 2025.
33. Hyuna Cho, Ziquan Wei, Seungjoo Lee, Tingting Dan, Guorong Wu, **Won Hwa Kim**, “Conditional Diffusion with Ordinal Regression: Longitudinal Data Generation for Neurodegenerative Disease Studies”, *International Conference on Learning Representations (ICLR)*, 2025. [Spotlight: ~5.1%]
34. Minjae Jeong*, Yechan Hwang*, Jaejin Lee, Sungyoon Jung, **Won Hwa Kim**, “HGM³: Hierarchical Generative Masked Motion Modeling with Hard Token Mining”, *International Conference on Learning Representations (ICLR)*, 2025. [*: equal contribution]
35. Hyeonwoo Cho, Chanmin Park, Dong-hee Kim, Jinyoung Kim, **Won Hwa Kim**, “CNG-SFDA: Clean-and-Noisy Region Guided Online-Offline Source-Free Domain Adaptation”, *Asian Conference on Computer Vision (ACCV)*, 2024.
36. Minjae Jeong*, Hyuna Cho*, Sungyoon Jung, **Won Hwa Kim**, “Uncertainty-aware Diffusion-based Adversarial Attack for Realistic Colonoscopy Image Synthesis”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024. [Provisional Accept: ~11%, *: equal contribution]
37. Seunghun Baek*, Jaeyoon Sim*, Guorong Wu, **Won Hwa Kim**, “OCL: Ordinal Contrastive Learning for Imputating Features with Progressive Labels”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024. [Provisional Accept: ~11%, *: equal contribution]
38. Yanquan Huang, Tingting Dan, Won Hwa Kim, Guorong Wu, “Uncovering Cortical Pathways of Prion-like Pathology Spreading in Alzheimer’s Disease by Neural Optimal Mass Transport”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024. [Provisional Accept: ~11%]
39. Yechan Hwang, Soojin Hwang, Guorong Wu, Won Hwa Kim, “Multi-order Simplex-based Graph Neural Network for Brain Network Analysis”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024.
40. Jaeyoon Sim, Minjae Lee, Guorong Wu, Won Hwa Kim, “Multi-Modal Graph Neural Network with Transformer-Guided Adaptive Diffusion for Preclinical Alzheimer Classification”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024.
41. Sooyeon Jeon, Yujee Song, **Won Hwa Kim**, “Gene-to-Image: Decoding Brain Images from Genetics via Latent Diffusion Models”, *MICCAI Workshop on PRedictive Intelligence in MEDicine (PRIME)*, 2024.
42. Wooseok Jung, Saehyun Kim, Dong-Hee Kim, **Won Hwa Kim**, “Modelling Brain Atrophy Dynamics Enhances Predicting Cognitive Decline in Alzheimer’s Disease Continuum”, *Cognitive Computational Neuroscience (CCN)*, 2024.
43. Tingting Dan, Mustafa Dere, **Won Hwa Kim**, Minjeong Kim, Guorong Wu, “TauFlowNet: Revealing latent propagation mechanism of tau aggregates using deep neural transport equations”, *Medical Image Analysis (MedIA)*, 2024.
44. Hyuna Cho, Jaeyoon Sim, Guorong Wu, **Won Hwa Kim**, “Neurodegenerative Brain Network Classification via Adaptive Diffusion with Temporal Regularization”, *International Conference on Machine Learning (ICML)*, 2024.
45. Tingting Dan, Ziquan Wei, **Won Hwa Kim**, Guorong Wu, “Exploring the Enigma of Neural Dynamics Through A Scattering-Transform Mixer Landscape for Riemannian Manifold”, *International Conference on Machine Learning (ICML)*, 2024.

46. Seunghun Baek*, Jaeyoon Sim*, Mustafa Dere, Minjeong Kim, Guorong Wu, **Won Hwa Kim**, “Modality-Agnostic Style Transfer for Holistic Feature Imputation”, *International Symposium on Biomedical Imaging (ISBI)*, 2024. [**Oral presentation**, *: equal contribution]
47. Yujee Song, Donghyun Lee, Rui Meng, **Won Hwa Kim**, “Decoupled Marked Temporal Point Process using Neural Ordinary Differential Equations”, *International Conference on Representation Learning (ICLR)*, 2024.
48. Inhyuk Park, **Won Hwa Kim**, Jongbin Ryu, “Style-KD: Class-imbalanced Medical Image Classification via Style Knowledge Distillation”, *Biomedical Signal Processing and Control*, 2024.
49. Jaeyoon Sim, Sooyeon Jeon, Injun Choi, Guorong Wu, **Won Hwa Kim**, “Learning to Approximate Adaptive Kernel Convolution on Graphs”, *AAAI Conference on Artificial Intelligence (AAAI)*, 2024.
50. Hyuna Cho, Yubin Han, Amal Isaiiah, **Won Hwa Kim**, “Covariate Correcting Network for Isolating the Impact of Long-term SES Changes on Brain Development”, *Annual Meeting of the Organization for Human Brain Mapping (OHBM)*, 2024.
51. Joonhyuk Park*, Yechan Hwang*, Minjeong Kim, Moo K. Chung, Guorong Wu, **Won Hwa Kim**, “Brain Connectome Analysis for Alzheimer’s Disease using Hodge Laplacian-based Edge Convolution”, *Annual Meeting of the Organization for Human Brain Mapping (OHBM)*, 2024. [*: equal contribution]
52. Hyuna Cho, Injun Choi, Suha Kwak, **Won Hwa Kim**, “Interactive Network Perturbation between Teacher and Students for Semi-Supervised Semantic Segmentation”, *Winter Conference on Applications of Computer Vision (WACV)*, 2024. [**First round accepted**: 92/815 = ~11%]
53. Hyuna Cho, Minjae Jeong, Sooyeon Jeon, Sungsoo Ahn, **Won Hwa Kim**, “Multi-resolution Spectral Coherence for Graph Generation with Score-based Diffusion”, *Neural Information Processing Systems (NeurIPS)*, 2023.
54. Tingting Dan, Jiaqi Ding, Ziquan Wei, Shahar Z Kovalsky, Minjeong Kim, **Won Hwa Kim**, Guorong Wu, “Re-Think and Re-Design Graph Neural Networks in Spaces of Continuous Graph Diffusion Functionals”, *Neural Information Processing Systems (NeurIPS)*, 2023.
55. Tingting Dan, Minjeong Kim, **Won Hwa Kim**, Guorong Wu, “Developing Explainable Deep Model for Discovering Novel Control Mechanism of Neuro-Dynamics”, *IEEE Transactions on Medical Imaging (TMI)*, 2023.
56. Hyuna Cho, Guorong Wu, **Won Hwa Kim**, “Mixing Temporal Graphs with MLP for Longitudinal Brain Connectome Analysis”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2023 [**Oral presentation**: 68/2250 = ~3%]
57. Hyuna Cho, Yubin Han, **Won Hwa Kim**, “Anti-Adversarial Consistency Regularization for Data Augmentation: Applications to Robust Medical Image Segmentation”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2023. [**Early accepted**: 14%]
58. Joonhyuk Park*, Yechan Hwang*, Minjeong Kim, Moo K. Chung, Guorong Wu, **Won Hwa Kim**, “Convolving Directed Graph Edges via Hodge Laplacian for Brain Network Analysis”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2023. [**Early accepted**: 14%, *: equal contribution]
59. Ellen Jieun Oh, Yechan Hwang, Yubin Han, Taegeun Choi, Geunyoung Lee, **Won Hwa Kim**, “REStoring Clarity: Unpaired Retina Image Enhancement using Scattering Transform”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2023.
60. Tingting Dan, Minjeong Kim, **Won Hwa Kim**, Guorong Wu, “Enhance Early Diagnosis Accuracy of Alzheimer’s Disease by Elucidating Interactions between Amyloid Cascade and Tau Propagations”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2023.
61. Tingting Dan, Minjeong Kim, **Won Hwa Kim**, Guorong Wu, “TauFlowNet: Uncovering Propagation Mechanism of Tau Aggregates by Neural Transport Equation”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2023.
62. Tingting Dan, Minjeong Kim, **Won Hwa Kim**, Guorong Wu, “Uncovering Structural-Functional Coupling Alterations for Neurodegenerative Diseases”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2023.
63. Jinhyeok Jang, Woo-han Yun, **Won Hwa Kim**, Youngwoo Yoon, Jaehong Kim, Jaeyeon Lee, ByungOk Han, “Learning to Boost Training by Periodic Nowcasting Near Future Weights”, *International Conference on Machine Learning (ICML)*, 2023.

64. Rui Meng*, Fan Yang*, **Won Hwa Kim**, “Dynamic Covariance Estimation via Predictive Wishart Process with an Application on Brain Connectivity Estimation”, *Computational Statistics and Data Analysis (CSDA)*, 2023. [Acceptance rate: ~13%, *: equal contribution]
65. Deunsol Jung, Sanghyun Kim, **Won Hwa Kim**, Minsu Cho, “Devil’s on the Edges: Selective Quad Attention for Scene Graph Generation”, *Computer Vision and Pattern Recognition (CVPR)*, 2023.
66. Huan Liu*, Tingting Dan*, Zhuobin Huang, Defu Yang, **Won Hwa Kim**, Minjeong Kim, Paul Laurienti, Guorong Wu, “HoloBrain: A Harmonic Holography for Self-organized Brain Function”, *Information Processing in Medical Imaging (IPMI)*, 2023. [**Oral Presentation**, *: equal contribution]
67. Seunghun Baek, Injun Choi, Mustafa Dere, Minjeong Kim, Guorong Wu, **Won Hwa Kim**, “Learning Covariance-based Multi-scale Representation of NeuroImaging Measures for Alzheimer Classification”, *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2023.
68. Injun Choi, Guorong Wu, **Won Hwa Kim**, “How Much to Aggregate: Learning Adaptive Node-wise Scales on Graphs for Brain Networks”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2022.
69. Tingting Dan, Hongmin Cai, Zhuobin Huang, Paul Laurenti, **Won Hwa Kim**, Guorong Wu, “Neuro-RDM: An Explainable Neural Network Landscape of Reaction-Diffusion Model for Cognitive Task Recognition”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2022.
70. Gangin Park, Chunsan Hong, Bohyung Kim, and **Won Hwa Kim**, “What Do Untargeted Adversarial Examples Reveal In Medical Image Segmentation?”, *MICCAI Workshop on Uncertainty for Safe Utilization of Machine Learning in Medical Imaging (UNSURE)*, 2022.
71. Xin Ma, **Won Hwa Kim**, “Locally Normalized Soft Contrastive Clustering for Compact Clusters”, *International Joint Conference on Artificial Intelligence (IJCAI)*, 2022.
72. Hyuna Cho, Gunwoong Park, Amal Isaiiah, **Won Hwa Kim**, “Covariate Correcting Network for Detecting Sole Effect of Socioeconomic Status on Brain in Children”, *Annual Meeting of the Organization for Human Brain Mapping (OHBM)*, 2022.
73. Hyuna Cho*, Feng Tong, Sungyong You, Sungyoung Jung, **Won Hwa Kim**, Jayoung Kim “Prediction of Response to Immunotherapy in Bladder Cancer Patients”, *IEEE Open Journal of Engineering in Medicine and Biology*, 2022. [*: Kim’s student]
74. Fan Yang, Guorong Wu, **Won Hwa Kim**, “Disentangled Representation of Longitudinal β -Amyloid for AD via Sequential Graph Variational Autoencoder with Supervision”, *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2022.
75. Hyuna Cho, Gunwoong Park, Amal Isaiiah, **Won Hwa Kim**, “Covariate Correcting Networks for Identifying Associations between Socioeconomic Factors and Brain Outcomes in Children”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2021.
76. Fan Yang*, Rui Meng*, Hyuna Cho, Guorong Wu, **Won Hwa Kim**, “Disentangled Sequential Graph Autoencoder for Preclinical Alzheimer’s Disease Characterizations from ADNI study”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2021. [*: equal contribution]
77. Xin Ma, Guorong Wu, Seong Jae Hwang, **Won Hwa Kim**, “Learning Multi-resolution Graph Edge Embedding for Discovering Brain Network Dysfunction in Neurological Disorders”, *International Conference on Information Processing in Medical Imaging (IPMI)*, 2021.
78. Debapriya Banerjee, Maria Kyrarini, **Won Hwa Kim**, “Image-Label Recovery on Fashion Data Using Image Similarity from Triple Siamese Network”, *Technologies*, 2021.
79. ByungOk Han, Woo-han Yun, Jang-hee Yoo, **Won Hwa Kim**, “Toward Unbiased Facial Expression Recognition in the Wild via Cross-dataset Adaptation”, *IEEE Access*, 2020.
80. Gowtham Krishnan Murugesan, Chandan Ganesh, Sahil Nalawade, Elizabeth M. Davenport, Ben Wagner, **Won Hwa Kim**, Joseph A. Maldjian, “BrainNET: Inference of Brain Network Topology using Machine Learning”, *Brain Connectivity*, 2020.
81. Tuan Q. Dinh, Yunyang Xiongy, Zhichun Huangy, Tien Voy, Akshay Mishray, **Won Hwa Kim**, Sathya N. Ravi, Vikas Singh, “Performing Group Difference Testing on Graph Structured Data from GANs: Analysis and Applications in Neuroimaging”, *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2020.

82. Fan Yang, Amal Isaiiah, **Won Hwa Kim**, “COVLET: Covariance-based Wavelet-like Transform for Statistical Analysis of Brain Characteristics in Children”, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2020. [Early accepted: ~13%, MICCAI 2020 NIH AWARDS]
83. Feng Tong*, Muhammad Shahid, Peng Jin, Sungyong Jung, **Won Hwa Kim**, Jayoung Kim “Classification of the Urinary Metabolome using Machine Learning and Potential Applications to Diagnosing Interstitial Cystitis”, *Bladder*, 2020. (*: Kim’s student)
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TEACHING EXPERIENCE

Instructor, Computer Science and Engineering, POSTECH, South Korea

- AIGS/CSED538: Deep Learning,
- AIGS/CSED526: Data Mining,
- CSED429F: Signal Processing

Instructor, Computer Science and Engineering, University of Texas at Arlington, U.S.A.

- CSE4334/5334: Data Mining,
- CSE6367: Computer Vision,
- CSE6363: Machine Learning

Teaching Assistant, Computer Sciences, University of Wisconsin - Madison, U.S.A.

- CS767: Computational Methods in Medical Image Analysis,
- CS638: Statistical Methods for Medical Image Analysis

Teaching Assistant, Robotics Program, KAIST, S. Korea.

- RE510: Intelligent Robot Design Lab.

SERVICES

<i>Associate Editor</i> , Pattern Recognition	2026-present
<i>Communication Chair / Local Committee</i> , MICCAI	2025
<i>Area Chair</i> , MICCAI, NeurIPS	2025-present
<i>Editor</i> , ICT Express	2021-present
<i>Senior Program Committee</i> , AAAI Conference on Artificial Intelligence (AAAI)	2021, 2026
<i>Reviewer</i> , MICCAI, ICML, NeurIPS, ICLR, CVPR, ECCV, ICCV, AAAI, ACCV, and WACV	2016-present
<i>Reviewer</i> , Medical Image Analysis (MedIA), NeuroImage, IEEE Transactions on Medical Imaging (TMI), IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), International Journal of Computer Vision, Applied Sciences, Neurobiology of Aging, and Alzheimer's and Dementia	2017-present

<i>Ad-hoc reviewer</i> , National Science Foundation (NSF)	2019
<i>Review Panel</i> , National Science Foundation (NSF)	2018

EXTRA ACTIVITIES

<i>Student Representative</i> , Robotics Program, KAIST, S. Korea	2009
<i>Volunteer</i> , International Federation of Automatic Control (IFAC), COEX, S. Korea	2008
<i>Volunteer</i> , International Workshop on Operator Theory and Applications (IWOTA), SNU, S. Korea	2006

PERSONAL REFERENCES

Available upon request.